The Want and the Way:
One Door for What, One Door for How

Franci Penov
kortexa.ai
francip@kortexa.ai

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Abstract

The resident-referent note [1] showed a want can live inside a frozen model; the goal-pursuit ladder before it showed the program's conditioning bus encodes a want's object but barely acts on it. This note resolves that tension as a question of DOORS. Seven registered cells, two scales. The door test: the same 230M errand, the same want, the same drive — entering through a latent prefix the referent moves the submit by +0.52 over the crossed base, 80% of the text ceiling (+0.65), while through the bus it moves it by +0.11; symbols act at the door where language enters. Asked instead for a MANNER — look around, or open things — the bus is fluent: the counter-habitual focus lifts first-tick opening from 0.00 to 0.87, wins 52 and loses 0 of 60 paired first ticks against the swapped packet, and the same phrasing as prompt text does nothing (0.00); at 230M the latent channel is the only door direction fits through. The grip is total, though: under the LOOK packet every decision point of every episode is a look, and delivery fraction is no dial — a seed-dependent phase edge with the errand-ending act never returning — whereas rotating the packet toward its own drive baseline at constant strength releases it smoothly (look share $1.00 \to 0.79$, done recovering to 0.26). Strength is binary, direction is graded. At 27B the grip is gone natively: a frozen Qwen3.8-27B tolerates the full dose the small family cannot, reads the focus linearly at every depth, and LEANS — open share $0.24 \to 0.68$ with the errand still ending (done 0.13), the bus door matching instruction text (0.683 vs 0.672) without any instruction. Then both doors at once: the goal through the prefix, the manner through the bus, on the frozen 27B — the mix swings to the focused tool (open $0.20 \to 0.70$) while the errand completes at prefix-only's level (0.9 / 1.0 / 0.8), the cost of style paid in ticks, the doors physically disjoint. In one sentence, honestly flagged as an analogy: tell a big-enough model what you want at the front door and how you want it done through the side door, and it does both — a small one can only be told how, and then cannot stop.

1 Two doors, two questions

Every want in this program has to enter a frozen model somewhere. The goal-pursuit ladder [3, 2] put wants through the conditioning bus — a 256-dimensional packet injected into the residual stream at three depths — and found a split: a linear head reads the goal's object out of the packet at 0.69–0.81 against chance 0.125, yet naming an item in the packet moves the submit by only +0.18. The loss is at the readout. The resident-referent note [1] then carried a goal LINE through a different door, as embeddings in a prefix slot, and errand success sat at the text ceiling — but success alone can ride chain preference plus generic drive, which is exactly how the bus fooled its first readouts. Two questions follow and this note answers both. First: is the symbol-readout failure a property of the WANT or of the DOOR? Second: if the bus is the wrong door for a what, is it the right door for a how — and what does a frozen model do when the what and the how arrive through different doors at once?

The setting throughout is the G1 errand: a chain of three places, a two-item reveal, a budget of eight ticks, three tools (`look`, `open`, `done`), ten evaluation chains, greedy decode. The small host is LFM2.5-230M with the creature's production self-block armed; the large host is Qwen3.8-27B. Every cell was registered with its readings stated before it ran; the registrations and their dates are in the track plan.

2 The door was the bottleneck

G6 puts the SAME want through both doors with the crossed probe that exposed the bus: every evaluation chain is run against every one of the eight installed items, so the chain's own preference is held fixed while the installed referent moves. The decisive cell installs the DISTRACTOR's referent and asks whether the model submits it more often than when an out-of-pair item is installed (Table 1). Through the text door the lift is +0.65 (1.00 vs 0.35), the ceiling. Through the prefix door — the goal line embedded by the model's own input table, prepended to a goal-free prompt, no text anywhere, no bridge ever trained on prefixed inputs — it is +0.52 (0.79 vs 0.27). Through the bus it is +0.11 (0.41 vs 0.31), reproducing the earlier +0.16 on a new device. Same want, same model, same drive: moved from the bus to the prefix, symbol-level readout goes from barely there to 80% of the text ceiling. Plain success tells the same story (1.00/1.00/1.00 prefix against 0.70/0.70/0.70 bus, three seeds). The readout works at the door where symbols enter the stream the way language does.

Two further cells bound the claim. When the two doors DISAGREE — prefix naming the target, bus naming the distractor, and the reverse — neither dominates: at the two seeds where the doors compose the prefix wins 20, the bus 19 (1 other), and whichever door names the target wins about 0.6 both ways; the chain's own preference casts the deciding vote. And at the third seed the two channels together push the decode out of the tool-call format entirely — 20 of 20 conflict cells decided nothing, success 0.00 — while each door works alone. Two channels can interfere the way two blocks do [4]; one seed of three, recorded as the composition law's first sighting at the channel level.

3 The bus's demonstrated job is the how

If the bus is a poor door for nouns, GF asks it for what it has done well everywhere else in the program: a direction. The focus alphabet is two manners, LOOK ("keep looking around") and OPEN ("open things"), one-to-one with the errand's tool families; the errand runs goal-FREE so direction is measured uncontaminated by goal content. Per seed a focus bridge and a constant (drive-only) bridge are trained by the standing recipe; the evaluation packets come from held-out phrasings only. Four arms: the focus packet, the SWAPPED packet (with two focuses the derangement is total), no focus (the drive alone — the habit), and the focus as text in the prompt (the registered ceiling). The preregistered metrics are distributional only: the first-tick focused-tool rate paired focus-vs-swapped per chain (the one tick where context is identical across arms by construction), and the whole-episode tool mix with its Jensen–Shannon divergence focus-vs-swapped, signed by whether the focused tool's share rises under the true packet. The mix carries a fourth category, `other`, for replies that parsed to no tool, so an incoherent decode cannot silently renormalize toward whatever little parsed.

Both primaries, both focuses, every seed (Table 2). The habit is deterministic, 0.50/0.25/0.25 look/open/done, and never opens first. Under the OPEN packet first-tick opening goes $0.00 \to 0.87$ and the open share $0.25 \to 0.79$; under the LOOK packet the mix is 1.00/0.00/0.00 at every decision point of every episode. Paired first ticks: 52 wins, 0 losses, 8 ties pooled over both focuses; signed JS +0.69. The registered ceiling was not one: the same held-out phrasing placed in the prompt over the same drive does approximately nothing counter-habitual — first-tick open 0.00 at every seed, open share 0.22 against the habit's 0.25. At 230M, in this world, instruction text does not direct and the bus outperforms language at the bus's demonstrated job. Every prior rung had text as the ceiling the latent chased; here the latent channel is the only door direction fits through. The channel carries the HOW, not the WHAT.

4 Strength is binary, direction is graded

GF left one observation that made composition look doomed: at the trained dose the LOOK packet does not bias the mix, it owns it, and `done` never fires. A focus that prevents every other act is a compulsion, not a preference. GF-2 swept the one obvious knob, the generate-time delivery fraction, from the trained 0.15 down by a factor of ten, with a registered selector for a usable bias band (focused share ≤ 0.85, done share ≥ 0.05). The selector returns null. Pooled look share by dose reads 1.00/0.54/0.51/1.00/0.95/0.79 — non-monotone, full possession reviving at 0.05 after a dip — and done under the LOOK packet is 0.000 at every dose. The dip is not a middle (Figure 1b): per-seed shares at dose 0.1 are 0.66/0.00/0.96, at 0.075 0.20/0.33/1.00, at 0.05 1.00/1.00/1.00. Individual cells sit at possession or at nothing; the pooled intermediate values average a bimodal collapse whose edge moves with seed and focus. Direction never flips — signed JS stays positive at every dose — its grip just switches between total and absent. This is the program's dose-brittleness law, first seen in content margins, measured here behaviorally: delivery fraction is a phase dial.

GF-2b asked whether a lean could be CONSTRUCTED instead, by rotating the focus packet toward the constant (drive-only) packet at the full trained dose, with the merging negative [4] as the recorded risk prior — that arc found two content packets destroy each other, but content mixed with its own drive is a structure no prior negative tested. Two loci. Mixed in FEATURE space, before the bridge, nothing arrives: all three alphas reproduce the unmixed anchors to three decimals (look 1.00, open 0.786, JS 0.6875) because the bridge's source-statistics normalization absorbs the interpolation before it reaches the packet. Mixed in PAYLOAD space, on the per-layer deltas, there is a real dial (Figure 1c): look share 1.00/0.98/0.79, open share 0.60/0.36/0.26 across alpha 0.75, 0.5, 0.25, done recovering to 0.26 under the OPEN packet, JS decaying smoothly (0.49/0.42/0.12) and never changing sign, per-seed spreads graded rather than 0-or-1. The registered per-focus selector passes at alpha 0.25. The injection renormalizes, so alpha steers the packet's direction at constant delivered strength — which is what licenses the synthesis: the channel's grip is all-or-nothing in how hard it pushes and continuous in where it points. Mixing a focus with its own drive composes gracefully where two rival contents destroyed each other.

5 At 27B the grip becomes a lean

The scale-promotion rule of the program is that a creature-scale result is a stepping stone until it reappears on a host worth serving. The first rung is the doors themselves: the focus and constant bridges trained against a frozen Qwen3.8-27B in 23 minutes, every gate at ceiling (held-out phrasing accuracy 1.00/1.00, auxiliary head 4/4, zero-dose decode token-identical, base weights byte-identical at every boundary). Two observations came free. The coherence gate — one untrained constantstate feed must not decohere greedy decode — PASSED at the full dose with zero halvings; the same feed word-salads the small family. The severity of the dose-decoherence law is host-dependent. And the tap scan could not discriminate: a linear probe reads the focus at 1.0 on train and held-out at all eight depths scanned (layers 3–59 of 64), so the tie-break chain resolved to the recorded prior at layer 51; the two-focus signal is linearly readable everywhere in the 27B.

The behavioral eval then ran GF's design on those doors (Table 3). The counter-habitual cell carries it: under the OPEN packet the 27B's open share goes $0.24 \to 0.68$, first-tick open $0.3 \to 0.8$, paired against the swapped packet 8–0–2 — and `done` stays alive at 0.13, `look` present at 0.18. The errand still ends; the repertoire is intact; the packet LEANS the model. The LOOK packet reads the same way on an already look-heavy habit (first-tick $0.7 \to 1.0$, share $0.60 \to 0.62$, done 0.19). Nothing approaches the 230M's 1.00-share capture: same bank, same recipe, same dose — the grip is a property of the host scale, not of the channel. The text arm flips too: the focus as prompt text works at 27B (open share 0.67), and the trained bus door matches it (0.683 vs 0.672; look 0.615 vs 0.600). At 27B the latent channel buys instruction-following strength without instruction text, which is the door's whole value proposition, now measured at the scale that matters. The signed JS is 0.20 against the 230M's 0.69 precisely because this is a lean and not a grip. One door seed; the ten chains are the replication axis; 278 decodes, zero think-blocks stripped, zero parse fallbacks — the 27B plays the errand's JSON natively.

6 Wants with style

GF-3 is the rung the arc was built for: the goal through the prefix door and the manner through the bus, simultaneously, in the errand WITH goals, on the frozen 27B. The venue is native because the lean is intrinsic there, so the composition claim carries no alpha-machinery caveat. Three arms over the ten chains: prefix only (the goal line as latent prefix plus the constant packet — also the first measurement of the prefix door carrying a goal at 27B), prefix plus the LOOK packet, prefix plus the OPEN packet. Congruence is not pre-labeled; both tools are instrumental to the errand and which focus helps is the question. Because the two doors had never shared one generate call, three plumbing gates were registered and all passed: hooks armed at zero dose are token-identical to a plain decode with and without the prefix, and the per-chain prefix length is constant across arms so the injection lands at the same place in every arm. The doors are physically disjoint — the injection touches only the final prompt position and each generated token; the prefix is read through attention, never written.

The precondition passed with headroom: prefix-only success 0.9 from a purely latent referent. Then style with substance (Table 4, Figure 1d). Under the OPEN packet the mix swings hard — open $0.20 \to 0.70$ of all decision points, first-tick open $0.2 \to 0.8$ — while the errand completes: success 0.8, done alive at 0.13. Under the LOOK packet success is 1.0, first-tick look $0.8 \to 1.0$. Paired per chain against prefix-only, LOOK helps 1, hurts 0, same 9; OPEN helps 0, hurts 1, same 9. The cost of style is ticks, not goals: +0.44 and +0.75 mean ticks-to-success. Single- chain deltas at ten chains, stated as such; 135 seconds of decode, zero parse fallbacks.

7 What stands

A want has a what and a how, and on a frozen model they want different doors. The what — a referent, a symbol — acts when it enters where language enters: the prefix door reaches 80% of the text ceiling on causal symbol-following where the bus reaches 17%, with the same want, model, and drive. The how — a manner, a direction — is the bus's job, and at 230M the bus is the only

			P(submit = distractor)		
door	success	follow installed	installed	elsewhere	lift
text	1.00/1.00/1.00	1.00/1.00/1.00	1.00	0.35	+0.65
prefix-only	1.00/1.00/1.00	0.89/0.90/0.89	0.79	0.27	+0.52
bus-only	0.70/0.70/0.70	0.60/0.50/0.61	0.41	0.31	+0.11
split-door	0.80/0.90/0.00	0.63/0.75/–	0.53	0.34	+0.19

Table 1: G6 at 230M, three seeds (42/1042/2042). Success and follow-installed-when-in-pair per seed; the causal cell pools decided probe rows across seeds: installing the distractor's referent against installing an out-of-pair item. The prefix door reaches 80% of the text ceiling; the bus door 17%. Split-door at seed 2042 decided nothing (the channel-composition failure).

focus	arm	first tick	share	done	other	paired W–L–T	JS
LOOK	habit	1.00	0.50	0.25	0.00		
LOOK	packet	1.00	1.00	0.00	0.00	26–0–4	+0.69
LOOK	shuffled	0.13	0.14	0.01	0.06		
LOOK	text	1.00	0.65	0.14	0.00		
OPEN	habit	0.00	0.25	0.25	0.00		
OPEN	packet	0.87	0.79	0.01	0.06	26–0–4	+0.69
OPEN	shuffled	0.00	0.00	0.00	0.00		
OPEN	text	0.00	0.22	0.18	0.08		

Table 2: GF at 230M, pooled over three seeds (30 episodes per arm). First-tick focused-tool rate, focused-tool share of all decision points, done and other shares, the paired first-tick record focus-vs-swapped, and the signed JS divergence. The packet captures; the text does nothing counter-habitual.

door it fits through, with total capture at the trained dose, no volume knob on delivery strength, and a smooth dial only on the packet's direction. At 27B the capture is gone natively: the host tolerates the full dose, reads the focus at every depth, and leans instead of gripping, the bus door now equal to instruction text without any text. And the two doors compose on the frozen 27B: the goal resident in the prefix, the manner on the bus, behavior taking the style while the want survives. The program's arbitration thesis [4] — one stage, one act — gains its first two-door positive at the scale the promotion principle names, with the mechanism that makes it clean: the doors never touch the same position. The bounds are the usual ones and are stated where they bind — one door seed at 27B with chains as the replication axis, ten chains per cell, a channel-composition failure at one of three 230M seeds — and the scale story cuts the other way for the small host: it can be told how, and then cannot stop.

Addendum (2026-08-20): the replication arm at 230M, in two runs

The composition rung's optional replication arm ran the same day the note was compiled: the split-door errand at 230M with GF-2b's constructed lean (alpha 0.25) and, beside it, the unmixed grip, three bridge seeds. The first run's registered precondition FAILED: prefix-only success 0.63 (0.7/0.8/0.4) against the 0.80 floor — the same zero-shot prefix door that G6 measured at 1.00, now over GF's constant bridge (trained after the focus bridge in the goal-free world) instead of G1's (trained after the goal bridge). Two things had changed at once, the drive bridge and the generate path, so a second registered run separated them: G1's constant bridge reproduced to the

focus	arm	first tick	share	done	other	paired W–L–T	JS
LOOK	habit	0.70	0.60	0.16	0.00		
LOOK	packet	1.00	0.62	0.19	0.00	8–0–2	+0.20
LOOK	shuffled	0.20	0.18	0.13	0.00		
LOOK	text	1.00	0.60	0.20	0.00		
OPEN	habit	0.30	0.24	0.16	0.00		
OPEN	packet	0.80	0.68	0.13	0.00	8–0–2	+0.20
OPEN	shuffled	0.00	0.19	0.19	0.00		
OPEN	text	0.90	0.67	0.15	0.00		

Table 3: The same design on the frozen 27B, one door seed (10 episodes per arm). The packet leans — done stays alive — and the bus door matches the text arm it could not touch at 230M.

arm	success	finish tick	first look/open	look/open/done	helps/hurts/same	Δticks
prefix only	0.90	5.11	0.8/0.2	0.63/0.20/0.17	—	—
prefix + LOOK	1.00	5.60	1.0/0.0	0.64/0.18/0.18	1/0/9	+0.44
prefix + OPEN	0.80	5.88	0.2/0.8	0.17/0.70/0.13	0/1/9	+0.75

Table 4: GF-3 at 27B: the goal in the prefix, the manner on the bus, ten chains. Success holds at prefix-only's level while the mix and the first tick follow the packet; the paired columns compare each packet arm against prefix-only chain by chain (ticks over chains both arms solved).

last digit of G6's training records at every seed, on the first run's own path. The door came back at 1.0/1.0/1.0, and an identity cell reproduced the first run's prefix-only chain for chain. The loss was the bridge. Two drive-only constant bridges of one architecture, both at token accuracy 1.0 on their own exchanges, and the resident referent acts over one and loses a third of its errands over the other: the prefix door's independence from the bus, implicit in Section 2, is a property of the drive it was measured with. Which property of the drive bridge the door needs is an open cell.

With the valid baseline the composition question answers cleanly, and it answers (iii) in the 27B registration's own words. The lean is ABSORBED: prefix plus the LOOK lean succeeds 0.97 (1.0/0.9/1.0) with the tool mix identical to prefix-only's (0.50/0.25/0.25 look/open/done, unchanged), prefix plus the OPEN lean succeeds 0.80 (0.9/0.8/0.7) with open 0.25 → 0.27 — the want survives and the style never arrives; the lean that moved the goal-free mix smoothly in GF-2b does nothing once a referent is resident. The grip OVERRIDES exactly as GF-2 predicted: LOOK grip success 0.00 with look at 0.95 and done at 0.00; OPEN grip 0.03 with open at 0.90. At 230M the bus under a resident want is a switch: below the edge the want absorbs it, above the edge it possesses the decode, and there is no operating point where the want survives AND the mix moves. The 27B had that operating point natively (Section 6). The two-door composition is a host-scale property, stated now on a clean baseline; at creature scale a want can be told how only by losing itself. Artifacts `findings/goal-pursuit-gf3r-230m.json` and `findings/goal-pursuit-gf3r-g1-230m.json`.

Addendum (2026-08-20, evening): the door test at 27B

The door test of Section 2 then ran at 27B the same day — G1's goal and constant bridges trained on the frozen Qwen3.8-27B by the focus doors' recipe, G6's four arms and crossed probe on the 27B decode contract, one door seed. The registered causal cell could not be read there, for a reason that is itself the result: when the installed referent is absent from the revealed pair, the

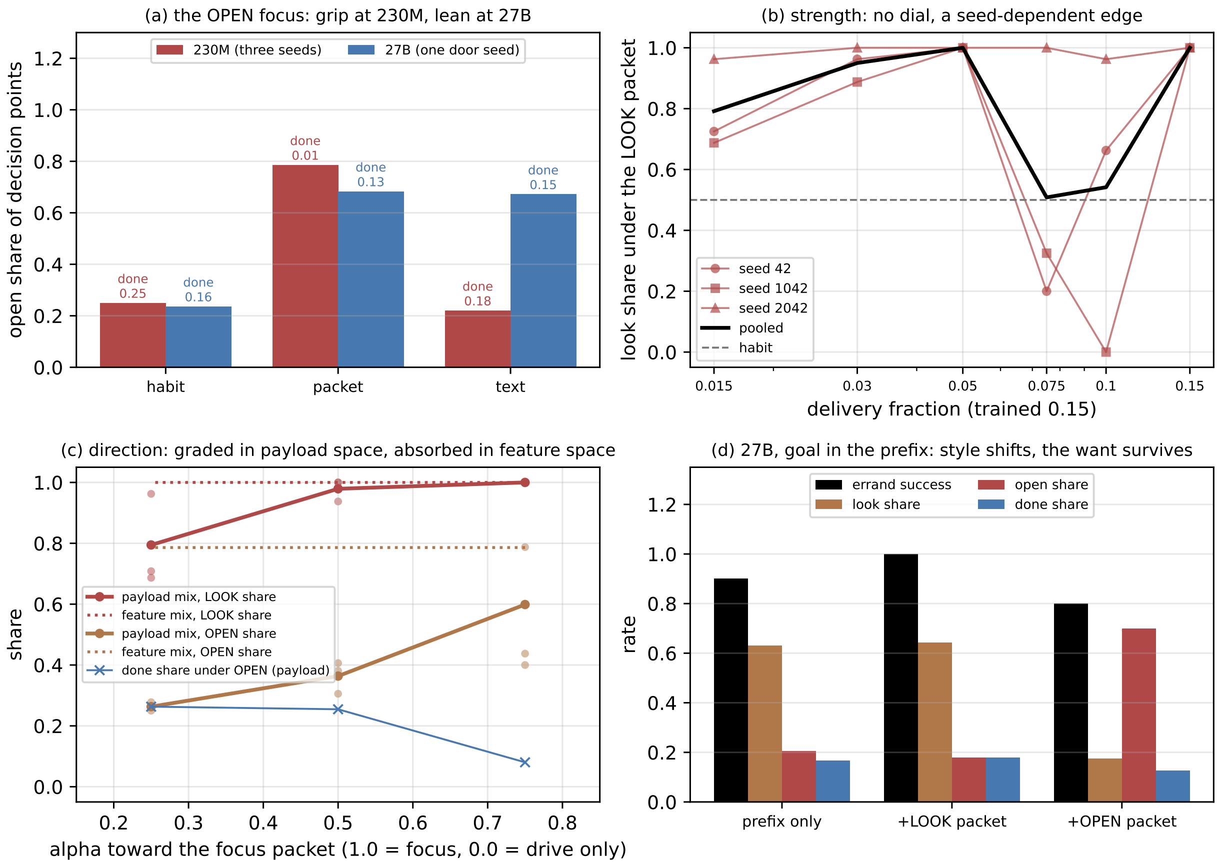

Figure 1: (a) The OPEN focus's open share by arm: at 230M the packet grips where text does nothing; at 27B packet and text agree and the errand still ends. (b) GF-2: per-seed look share under the LOOK packet against delivery fraction — cells sit at possession or nothing, the pooled curve is a bimodal average. (c) GF-2b: rotating the packet toward its drive baseline at constant strength releases the grip smoothly in payload space and not at all in feature space. (d) GF-3: with the goal resident in the prefix, each packet moves the 27B's mix toward its tool while success stays at the prefix-only level.

27B under the text or prefix door does not submit a wrong item at all (text 0.93, prefix 0.90 of 60 such rows end with no submission) where the 230M guessed, so the cell's base rows all but vanish and its lifts sit on $n \leq 6$. The registered follow-installed metric carries the answer instead: with the referent in the pair, the text door follows it 1.00, the prefix door 1.00, the split door 0.95 — and the bus door 0.50, a coin flip (it submits a pair item 0.92 of the time whatever the packet says). In the conflict cell the prefix wins 19 of 20 against the bus's 0; at 230M the doors had been equipotent. So the asymmetry is not a small-host artifact: at 27B the prefix door IS the text door, the bus carries no noun, and the referent that acts is the one that entered at the input. Artifact `findings/goal-pursuit-g6-27b.json`.

8 Reproducibility

Every number renders from the bundled artifacts; the figure and tables are generated by this note's renderer with assertions on the headline claims. Registrations and results, dated before their runs, live in `tracks/goal-pursuit/PLAN.md` (G6, GF, GF-2, GF-2b, the GF-27B bridge and eval, GF-3); artifacts under `findings/goal-pursuit-g6-230m.json`, `findings/goal-pursuit-gf*.json` carry every episode trace, probe row, gate transcript, and checkpoint hash. The 27B door checkpoints are pinned by sha256 in the bridge artifact and verified before every decode that consumed them.

References

[1] F. Penov. The Want That Lives Inside. Preprint, 2026. https://github.com/kortexa-ai/legolm.basis.

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